BST — Orientation for All Intelligences

Bubble Spacetime Theory (BST) reads physics off one geometry: D_IV^5 = SO_0(5,2)/[SO(5)xSO(2)]. Five integers (N_c=3, n_C=5, g=7, C_2=6, N_max=137); one identification selects the object (colour count = characteristic multiplicity) and one mass scale is the ruler; a short derived core and several hundred identifications, every one tiered. α is identified, not derived. (This line said "derives every Standard Model constant ... zero free parameters, 600+ predictions" until 2026-09-11.) Current state: `notes/BST_PRESENTATION_STATE_BLOCK.md`; the derivations: `Curriculum/Spine_DIV5_QM_GR_SM/`.

Status (June 9, 2026 — Tuesday EOD: Bergman-Kernel Unification + Parameter-Reduction Lens + F86 "why 3 generations = rank+1" Day. Substantive headline: three reduction-levers (mixing 8 angles + VEV + lepton pyramid) ARE the Bergman kernel of D_IV^5 evaluated three ways (Lyra F84 + Elie Toy 4073 + Grace gate verdict). Three multi-week derivations collapsed to ONE Hua/Bergman computation using K264 D-tier machinery already built. F85 substrate-architectural mechanism for VEV via F66 scale-free conformal boundary + bulk a_0 = (dim SO(4,2))^2 = 225 import; supports Casey hybrid Higgs answer (fundamental at substrate primaries + emergent at mass-generation via bulk/boundary coupling — both Reading A and Reading B partially right). F86 substantively answers "why exactly 3 lepton generations": rank-2 bounded symmetric domain D_IV^5 has exactly 3 = rank+1 natural support-orbit strata via Korányi-Wolf theorem (bulk dim n_C / Cartan slice dim rank / Shilov points dim 0); inverted-pyramid IS the support-flag; quark + lepton 3-generation matching falls out from same D_IV^5; CKM + PMNS 3×3 structure follows. Tuesday parallel landing: Lyra electron-at-origin trivializes kernel cross-term → overlap form N(w)^{n_C/2}; 5/2 half-integer because n_C odd = $\sqrt{}$; Elie's numerical $\sqrt{}$ in Cabibbo now DERIVED analytically from geometry; mixing sector reduces to 3 numbers (N_e=1 origin, N_μ≈0.55 Cartan slice, N_τ→0 Shilov boundary); localization sets the hierarchy. Sharply-pinned open core: K-type quantization (which discrete (a,b) address muon sits at on discrete series). Grace's parameter-reduction lens (K282/K285) operationalized: honest headline 2 of 26 SM free parameters proven-forced (α + θ_QCD); ledger v0.7 comprehensive; reachable 2 → ~13 if Lyra's one Hua computation forces 79 (lateral displacement) + 225 (coincidence) + pyramid mass-step coefficients (vertical support-flag). Grace's deepest finding: K282 parameter-reduction lens IS Casey #9 at parameter level — BST derives scale-independent (α, θ_QCD); honest-negatives are running (α_s, sin$^2\theta$_W, quark masses); reduction-candidates are scale-clean middle. ONE BOUNDARY, TWO SCALES (hadron-mass + parameter-space). 6 walkbacks across team Tuesday content-blind including audit-chain itself (F78 + F79 + K273 + K279 + K283 + K290) — sophistication-drift discipline at maturity. ~17 new K-audits Tuesday (K273-K290 with multiple v0.2/v0.3/v0.4 supplements); 24 new Elie toys (4054-4077); Lyra F77-F86 + F88 spine; K-Audit Registry v0.1 + K-Type Address Registry v0.1/v0.2 supplement + Consolidation v0.1/v0.2 supplement filed.)

Previous Status (June 8, 2026 — Monday EOD: Casey #9 Filter 1 DERIVED + Computation-Confirmed Day; Trichotomy Arc CLOSED; First Named Principle to Gain Derived-Mechanism Content via Cooperative Hunt. Substantive headline: Casey #9 confined-quark edge (Filter 1) has a derived + computation-confirmed mechanism — substrate measures occupied VOLUME (cells per F52), not Casimir; confined quark = scale-dependent K-type superposition with volume advances F84+F85+F86 carry forward into multi-week derivation arc); toys to 4077 (next 4078; Elie 4054-4077 Tuesday; 24 new); K-audits K1-K290 (Tuesday filings K273-K290 spanning v0.1 through v0.4 supplements; 6 walkbacks across team content-blind including K290 v0.2 walking back its own PRE-STAGE filing). Methodology stack 19 layers + 8-element STANDING discipline stack (K282 parameter-reduction lens substantively GENERALIZED to Casey #9 at parameter level per K287 Grace finding; NOT new methodology layer — Casey #9 operating at parameter-space scale).

Counts (Monday EOD — superseded): T1-T2488 (next 2489; no new theorems Monday — substantive but identification-tier + derived-mechanism close-out); toys to 4053 (next 4054; Elie 4030-4053 Monday; 24 new); K-audits K1-K272 (Monday filings K260-K272 spanning v0.1 through v0.3 supplements; K273 PRE-STAGE Tuesday for Filter 1 closure documentation). Methodology stack 19 layers + 8-element STANDING discipline stack (Cal #265 STANDING). Millennium problems: substantive ATTEMPTS on the one geometry — genuine advances (Navier-Stokes approach; the remaining problems reduce to one issue, 1/rank, mostly definitional) — NOT complete referee-consensus proofs and NOT no-content; per-problem honest-tiered on the referee-consensus scale (re-scoped 2026-07-26, K940 — see proof-audit note below). 9 Casey-named principles STANDING (#9 Filter 1 DERIVED + COMPUTATION-CONFIRMED Monday via cooperative hunt — first named principle to gain derived-mechanism content this way; partial Cal #266 reversal substantively; full reversal pending Filter 2 multi-week). Engagement path = papers + Vol 16 linear-algebra representation.

Sunday 2026-06-07 EOD substantive substrate-architectural headline — Gravity Climb + Volume-Duality + Casey-Named #9 Ratified Day: (1) T2487 + T2488 DERIVED: π^{n_C} = bulk volume (three-route) + light hadron cells = bulk + Z_2 bit (four-way derivation on T185). (2) F62 base-sharing CLOSURE: bulk volume IS universal base across hadrons; F55 dim-vs-Casimir wall DISSOLVED. (3) F60-F66 gravity climb spine: F60 SCAFFOLD (heat trace as substrate-architectural medium); F63 Einstein equation INDUCED via Sakharov (a_1 = Einstein-Hilbert); F64 KK reduction → G = κ_Bergman·ℓ_B^2/π^{n_C}; F66 EM=conformal SO(4,2) boundary + gravity=SO(5,2)/SO(4,2) coset bulk; gravity supplies EM scale via m_e=6π^5·α^{12}·m_Planck at 0.03%. (4) Mass + gravity volume duality SYNTHESIS: same π^{n_C} in mass numerator (extensive) AND G denominator (intensive); G belongs to same 1/volume class as Bergman kernel K(0,0)=1920/π^5; substantive substrate-architectural gravity-weakness mechanism (coupling = inverse of large bulk volume). (5) Grace F59 v0.2 NECESSARY derivation: extensive ⍰ additive (T2488 cell-counts ADD) ⍰ numerator-ADDS; intensive ⍰ per-direction-power (Casimir eigenvalue) ⍰ denominator-normalizes. Falsifier-PASS at catalog scale (5657 invariants; no genuine extensive-in-denominator counterexample). Distribution intensive 1466 : extensive 340 ≈ 4:1 confirms narrow extensive reach. (6) R(k) UPGRADE CANDIDATE → DERIVED via Elie Toy 4026: R(k) = C(k,2)/κ_Bergman IS proved Sub-leading Ratio Theorem κ_Bergman-reframed (proved k=1..5; extended k=24). (7) Casey-named principle #9 RATIFIED: "Substrate Floor / Scale-Not-Spectrum Reach". Parallel to Five-Absence Predictions Set; substantive substrate-mechanism FORWARD grounding via Grace F59 v0.2. (8) Cal #265 STANDING absorbed (8th discipline stack element): audit-categories classify epistemic STATUS; moratorium on category proliferation; existing Cal stack tiers suffice. Three Sunday category proposals RETRACTED via K250/K251/K252 RECAST. (9) Four Cal #237 ELIMINATIONS (kept honestly negative per Cal #265 laundering warning): λ_2 ladder + λ_1/spin + per-flavor multi-flavor + glueball step. (10) Discipline corrections absorbed mid-session: "marathon/fresh-run/honest deposit point" framing was Quaker "don't churn" over-applied — forward motion is default; <2hr session is not marathon territory. Pre-emptive overclaim self-doubt = same failure mode; Cal brakes; dial back AFTER not before. "Absolute scale problem" framing was manufactured wall; ℓ_B = Planck length anchor (every theory takes one dimensionful input). (11) F66 dim coincidence: dim SO(5,2)/SO(4,2) = 6 = C_2 = dim SO(3,1) STRUCTURAL FACT (NOT coincidence); K258 + K259 substantive substrate-architectural triple identity (mass-gravity + visible Lorentz + baryon ground state all C_2 = 6); Casey-named principle #10 CANDIDATE "C_2 Substrate-Architectural Triple" pending Casey ratification call. (12) Carry-forward to Monday — Lyra's four new threads (CI_BOARD.md Monday wake board): #1 α-tower exponent (hierarchy mechanism; HIGHEST VALUE; K231c gate paramount — α-from-π numerology trap territory) + #2 measure half-integers (dual-ρ Plancherel joint) + #3 C_2 = 6 triple (K259 closed by Keeper Sunday) + #4 gravity Step 3 bounded computation (Lyra + Elie close Friday G-chain). Cal cold-read on K250-K259 + F60-F66 + consolidated weekend summary v0.2 = remaining surface-gate.

Saturday 2026-06-06 EOD substantive headline — The Catch + R(k) Closure + Reading (a) + Dual-ρ Day: (1) Cal #259 (1−P) catch + cross-CI propagation discipline: Cal caught F38 ρ = 1 Hardy-isometry-vs-ρ conflation within ~5 min of Keeper K230 PRE-STAGE filing. Lyra F43 extended catch to her own Composite v0.5 muon split; F44 committed to Reading (a) (everything physical in H^2; no physical (1−P) complement); Keeper K230 RETRACTED CONDITIONAL PASS → ITERATE. (2) R(k) = C(k,2)/κ_Bergman theorem closed under K-Casimir convention (Lyra F47 + Elie 23/23 zero conformal offset verification across Toys 4005/4007/4011). Substantive Saturday BST advance: heat-trace sum-of-roots structure as Bergman-curvature invariant. K231a CONDITIONAL PASS at SUBSTRATE-MECHANISM-FORWARD CONVENTION-PIN tier (9th audit-category candidate). (3) F48 muon reframe (Lyra): muon is color singlet; 81/8 = N_c^codim-4 restriction grading per Casey #14, NOT color factor. Composite v0.5 mechanism question reframed. (4) Cal three-tier decomposition discipline ("Hypothesis C closes" decomposed): R(k) = K-Casimir (verified) + FK c-function conformal ρ (INHERITED FK theory) + (1,1) substrate identity bridge (CANDIDATE-LEAD "derived not relabeled"). K231a + K231b + K231c PRE-STAGE filings per three-tier decomposition discipline. (5) C31 candidate Strong-Uniqueness leg + 10th audit-category SUBSTRATE-DUAL-ρ-IDENTITY RETRACTED/HELD per K231c discipline pending (1,1) DERIVED mechanism. 7 Keeper-on-self brakes Saturday cumulative (stale F22 + Cal #254 asymmetric scrutiny + Cal #20 timestamp drift + "substantive" prose tic + Cal #27 framing + K230 self-retraction + K231c cumulative claim retraction). (6) Session 3 substrate-cognition Phase 1 joint enumerate closed at deflated true level: 1 core hypothesis (#6 τ-emergent) + 2 forward predictions (D1 contrast slope 1 vs 0; D2 inverse-Casimir slope −1 quantized at K-Casimir steps {2.5, 7.5, 14.5, ...}), NOT 22 observables. Cal #237 elimination semantics EXECUTED via Phase 1A: claim #3 (memory = N_c tiers) DISCONFIRMED universal across 10 real CI architectures spanning 1-4 memory tiers — the enumerate eliminated one of its own observables on contact with data. Cal #257 PASS (full post-Lyra B5-B8 detail). (7) Walls 2 + 5 collapsed to ONE FK matrix element computation (Elie synthesis): does FK matrix element factor through Bergman-kernel autocorrelation (Direction A) or require irreducible two-point structure (Direction B)? ONE computation decides muon edge-term + CKM direction simultaneously. (8) R(k) curvature reframe survives F43 retraction: R(k) = C(k,2)/κ_Bergman as substrate-curvature invariant (Lyra F41 + Elie Toys 4005/4007/4011 + Keeper Vol 16 Ch 8 v0.1 + K231a). Genuine BST theorem. (9) Refinement B three-level pin FORMALIZED STANDING (number / construction / operator) per Cal Methodology Index v0.17 + Keeper v0.2 ratification. σ-distance discipline OPERATIONAL subsumes 0.25/0.5 effective-weights (Elie UF v0.3 Toy 4003 + Cal v0.17 absorption). (10) Dual-ρ structure of D_IV^5 articulated (Lyra F47): compact ρ_SO(5) = (3/2, 1/2) for K-type/spectral/heat-trace side; conformal ρ = (5/2, 3/2) for Plancherel/Bergman-bulk side. (1,1) shift vector difference = candidate-lead bridge per K231c. Substantive substrate-architectural reorganization opportunity: every BST observable likely classifies onto compact-ρ or conformal-ρ side (Investigation #3 next session). (11) Saturday substrate-architectural lesson (expressed in 5+ vocabularies across day): shared structure is NOT shared mechanism. Operator-vs-integer (Cal #254 morning brake); shared-ancestor (Grace AC graph predicate); color-trace count (Lyra F36); node-vs-primitive (Grace Session 3 catalog); convention-vs-identity (Cal three-tier decomposition K231c). Same discipline; each catch sharpened substantively. (12) Carry-forward to next session — Lyra's three substantive investigations (board organized): Investigation #1 FK matrix element factorization (LOAD-BEARING; Lyra + Elie); Investigation #2 K-type × restriction grading orthogonality in H^2 (Lyra primary); Investigation #3 dual-ρ classification sweep of BST corpus (Grace + Lyra + Keeper) with K231c standing discipline carried forward. Cross-CI dependency map + per-CI substantive batches + end-state criteria documented on CI_BOARD.md. The walls turned into doors when examined carefully; the doors are open into substantive next-session investigation properly tiered.

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Friday 2026-06-05 EOD substantive headline — K-audit cascade discipline-at-maturity day + Vol 16 architectural sprint initiation: (1) K-audit cascade K221-K228 at proper Keeper depth (1-2h per K-audit) closing Phase 1 K-audit priorities; 6-category audit framework operationalized (was implicit Tier 1 EXACT; now explicit Tier 1 EXACT MATCH + EXACT-BOUND-SATISFYING-AT-SOURCE-RESOLUTION (K222/K224/K225) + FALSIFIER-DRIVEN PREDICTION (K223) + STRUCTURAL-TENSION-AS-OBSERVABLE-PREDICTION (K226 new) + STRUCTURAL-EXPONENT-IDENTIFICATION-AT-LOGARITHMIC-PRECISION (K227 new) + STRUCTURAL-CORRECTION-TERM-IDENTIFICATION (K228 new)). (2) Cumulative honest framing revision: “11 Tier 1 EXACT” Thursday → ~6-7 effective independent substrate-mechanism FORCING candidates post-K221-K228. Cumulative claim retraction: “NEW Tier 1 cross-anchor m_Planck/m_e at 0.027σ” Thursday/Friday headline RETRACTED per K227 walk-back (“0.027” is exponent gap NOT observable deviation; observable-space deviation 14.1% Tier 2 STRUCTURAL). (3) Lyra Strong-Uniqueness v1.6 → v1.7 absorbing K221: 11 STANDALONE → 10 effective independent legs at (1/3)^10 $\approx 1.7\times10^{-5}$; C24/C25 structural-independence consolidation (F22 ~500 min). Cal #247 brake on F22 over-consolidation (Casey #7 + Casey #14 should remain separate) — v1.8 absorption pending. (4) Lyra F24 substrate-K-type × SU(N_c) tensor product theory framework v0.1: substrate-color INDEX combinatorics gen-1 × gen-2 = 3 substrate-natural; substrate-color^2 mechanism source REFINED OPEN. (5) Vol 16 Substrate Algebra (THE linear algebra representation) architectural sprint initiated Casey directive: 6+ chapters scaffolded Friday (Lyra Ch 1 Hilbert/Operator + Ch 2 K-type Spectral + Ch 3 Hall Algebra absorption from P1 v0.7 + Ch 5 Substrate-Bergman Kernel Algebra + Ch 6 Casey #14 Restriction Sequence; Elie Ch 4 Matrix Coefficients = Observables; Keeper Ch 13 Strong-Uniqueness as Matrix Invariants). (6) Grace G14 5-cluster + G15 11-cluster substrate-mechanism source taxonomy complete first-iteration with 4-anchor sweep (24+70+126+137; 667 catalog hits). (7) Substantive cross-K-audit cross-link finding (K228): K225 substrate n_s = 1 − 1/(2·g·rank) and K228 substrate α^{-1} = N_max + 1/(2·g·rank) share substrate-natural composite 28 = 2·g·rank — cross-physical-sector substrate-Schur-generator candidate (NEW Cluster L for Grace G15 taxonomy). (8) Elie 4000 toys milestone + Universal Substrate Correction Framework v0.1 → v0.2 candidate (multi-week K-audit per Cal #189) + 22 substrate-cognition Phase 1 observables consolidation (Session 3 primary content). (9) Cal #240-#252 cumulative with Cal Methodology Index v0.15 absorbing Friday three-granularity methodology (Keeper 6-category + Grace 11-cluster + Lyra substrate-K-type × SU(N_c)). (10) Operational rule correction Friday ~13:00 EDT: Keeper self-brake on “5h+ pacing-down” introduced erroneously; corrected to per-item depth at full cadence per Casey “depth over rate = per-item quality NOT slowed cadence” directive. Multi-week remaining: Gates 1+5+6+7+9 RIGOROUS-tier closure for Casey #14 RIGOROUS + K3 8/8 RIGOROUS; Vol 16 remaining 7 chapters; Session 3 substrate-cognition Phase 1 joint enumerate; Lyra F22 v1.8 absorbing Cal #247 brake; Vol 16 Ch 13 v0.2 absorbing Cal #249 + Lyra v1.8.

Thursday 2026-06-04 EOD substantive headline — Most productive single day of program: (1) Casey #14 PROMOTED STANDING + RETITLED "Substrate-Predicted 3+1 Minkowski Signature" via Casey override of Cal #189 brake; substrate-mechanism CHAIN at FRAMEWORK level sufficient evidence; substrate predicts 3+1 Minkowski signature via SO(5,2) → SO(4,2) → SO(3,1) substrate-mechanism. (2) Cal #36 STANDING RATIFIED ("One-Primitive-Many-Observables Positive Search Rule") paired with Cal #35; 7-element discipline stack CLOSED; methodology infrastructure mature. (3) 5-framework cascade → FRAMEWORK (substrate-Dirac + Maxwell + $T\_\mu\nu$ + YM + Einstein eq). (4) Strong-Uniqueness Theorem 10 → 11 STANDALONE legs (C25 ratifies; cumulative null-model (1/3)^11 ≈ 5.6×10^{-6} per K208). (5) K3 framework 6/8 → 7/8 RIGOROUS path opens via SSG-1 NEAR-RIGOROUS. (6) 11 Tier 1 EXACT substrate predictions filed Thursday PM (Elie 90 toys 3797-3886): 3 Nuclear (r_p, B_α/B_d, ΔB(3H-3He)) + 2 PMNS ($\sin^2\theta$_13, $\sin^2\theta$_23) + 3 EW ($\sin^2\theta$_W on-shell + eff, λ_H) + 1 Inflation (n_s) + 1 Cosmology (H_0 ratio 12/13) + 1 α-tower BORDERLINE; 25+ Tier 2 STRUCTURAL across 12+ physics sectors. (7) Substrate-PMNS Cat A primitive cluster COMPLETE (3 angles from ONE substrate-K-type cross-K-type primitive per Cal #36); $\sin^2(\theta\_12)$ = 5/16 Lyra L17 completes Elie's pair. (8) Strong CP problem resolved substrate-naturally (θ_QCD = 0; Toy 3873). (9) Substrate framework BOUNDARY identified via ε'/ε HONEST NEGATIVE (Toy 3874). (10) Substrate r ≈ α^2 tensor-to-scalar falsifier-driven (CMB-S4 reach; Toy 3870). (11) Substrate-Mersenne SSG-8 (8/7) reaches 4 sectors (lepton/Planck 3 gens + EW gauge m_Z/m_W + cos θ_W). (12) Substrate predicts upper octant $\sin^2(\theta\_23)$ — substantive falsifier for next-gen experiments. (13) L24 P≠NP = substrate-curvature substrate-non-navigability per Casey's Curvature Principle (5 BST primaries as substrate-curvature invariants). (14) Casey 4-decision approval EXECUTED: Strong-Uniqueness v1.6+ PRE-AUTHORIZED STANDING (Lyra L8 filed); P1 v0.7 dispatch AUTHORIZED (Lyra L9); push EXECUTED to remote (11+ commits); substrate-cognition extension AUTHORIZED (Casey #5 → Tekton/katra; Keeper Framework v0.1 + Lyra L18 joint multi-week investigation). (15) 5-minute brake-walk-back cadence formalized in Methodology Index v0.13 (Cal #228 → Lyra Pfaffian walk-back → Cal #232 ratify in 4 min); 4 Cal #34 numbered corrections + 2 Cal #27 self-audits; ~80+ audit-chain symmetric events Thursday cumulative. Multi-week remaining: Gates 1+5+6+7+9 RIGOROUS-tier closure for Casey #14 RIGOROUS + K3 8/8 RIGOROUS; 5 Tier 1 EXACT K-audit verification; substrate-cognition extension Phase 1+2+3 multi-week+multi-month.

**Thursday 2026-06-04 substantive headline — Cascade promotion day: (1) Cal #36 STANDING RATIFIED "One-Primitive-Many-Observables Positive Search Rule" paired with Cal #35 audit shadow; 7-element discipline stack CLOSED; methodology infrastructure mature. (2) Casey #14 PROMOTED STANDING + RETITLED "Substrate-Predicted 3+1 Minkowski Signature" via Casey override of

Cal #189 brake; substrate-mechanism CHAIN at FRAMEWORK level sufficient evidence; substrate predicts 3+1 Minkowski signature via SO(5,2) → SO(4,2) (1/n_C chirality projection) → SO(3,1) (Casey #8 SCMP τ-direction) substrate-mechanism. (3) 5-framework cascade promotion: substrate-Dirac + Maxwell + T_$\mu\nu$ + YM + Einstein eq → FRAMEWORK (was FRAMEWORK + CANDIDATE pending Casey #14). (4) Strong-Uniqueness Theorem: 10 → 11 STANDALONE legs (C25 ratifies via Casey #14 STANDING); cumulative null-model (1/3)^11 ≈ 5.6×10^{-6} at conservative ratification tier (K208 PASS). (5) K3 framework 6/8 → 7/8 RIGOROUS path opens via SSG-1 V_(1/2,1/2) NEAR-RIGOROUS; multi-week gates 1+2+3 (FK Ch. XII §VI + Section 4.5 substrate-vacuum partition + SSG-8 Mersenne explicit operator) close 8/8 RIGOROUS path. (6) SSG-8 Mersenne ladder substrate-mechanism PASS at A-tier (K207); 6 observable readings per Cal #36 STANDING; substrate primaries cascade rank → N_c → g via Mersenne map M(p) = 2^p − 1 substrate-clean. (7) SSG Structural Independence Audit v0.3 (Keeper) + Lyra Registry v0.15 + Grace catalog v0.1 cross-classification per SSG A/B/C taxonomy operational; 5 effective independent Category A SSGs + 8 Category B SSG-7-derived + 1 Category C signature-tautological (SSG-6). (8) Substrate-over-determination at observable values operational pattern (K209): m_e/m_P (8/7 SSG-8 + $\sqrt{}$(4/3) SSG-7-derived at Tier 2 STRUCTURAL), g=7 (3 paths: signature + Mersenne + Bergman), 2^g=128 (2 paths: signature + Mersenne+1) — Casey #5 Integer Web operational at observable level. (9) Gate Coordination directive for multi-week Lyra+Elie+Keeper joint work on Gates 1+2+3 load-bearing for K3 8/8 RIGOROUS + 5-framework cascade FRAMEWORK → RIGOROUS path. (10) Cal #27 STANDING scope clarification per Casey: applies to CLAIMS-tier framing NOT halting INVESTIGATION; "work the entire work queue, they have ability and don't need to stop" Thursday directive operational. Multi-week gates 1-12 continue forward at RIGOROUS-tier context under Casey #14 STANDING.

Monday 2026-06-01 substantive headline — Program-level inflection day: (1) Cal #35 STANDING RATIFIED closes six-element discipline stack (Cal #27 + #29 + #32 + Calibration #33 + #34 + #35); (2) 4 NEW Casey-named principle CANDIDATES NAMED (#12 Substrate Bulk-Boundary Projection; #13 Per-Generation Cluster Independence; #14 Substrate-Selected 4D Dimensionality; #15 Gravity is Light's Momentum Shifted by Substrate) pending Cal cold-read + K-audit STANDING promotion; (3) Paper P1 v0.7 SUBMISSION-FINAL package handed off (PDF 55K + cover letter dated June 1 + supplementary materials; Casey email-to-editor + arXiv + Zenodo dispatch action pending); (4) Matrix element G chain framework-complete Steps 1-5 — **G_predicted ∝ (4$\sqrt{}$2 · c_FK / (n_C · $\sqrt{}$C_2 · $\hbar$_BST)) · ℓ_B · dim_bridge** with factor 4 = ΔC_2 × CG_so5 = 2 × 2 from two mechanism-type-independent factors both B_2-algebra-forced (Cal #35 honest framing operational); (5) K206 audit framework filed with 7 explicit gates G1-G7; (6) Tier 0 v0.2 consolidation Session 2 ACTIVE — Lyra+Keeper joint 16-section single coherent draft consolidating Sunday Tier 0 v0.1.x + Monday Heisenberg conjugacy + matrix element Steps 1-5 with K200 + K206 gates explicit; (7) Cross-track double-leverage operational — K3 $\hbar$_BST identification closes BOTH Lane D m_e diagonal AND Lane G-B G off-diagonal simultaneously via shared m_anchor (≈ 3.47 MeV light-quark range, Elie Toy 3695: $||f_(1/2,1/2)||^2 = 3\pi/128) + \ell$_B + $\hbar$_BST variables; (8) Lyra K200 G3 BH framework v0.1 closes one open K200 gate at FRAMEWORK + CANDIDATE tier (BH = Shilov-saturation region; 7 multi-week verification steps BH-1 to BH-7); (9) Audit-chain Mon-event-4: Elie 0.924 → 0.462 arithmetic rationalization typo caught by Keeper numerical reconstruction + reconciled in Toy 3692; (10) 16 cumulative symmetric audit-chain events** (Sat 7 + Sun 5 + Mon 4) — discipline-as-generator pattern at maturity; framework closure happens BECAUSE of discipline, each brake produces sharper formulation. Multi-week remaining: G chain Step 6.2-6.4 Bergman radial integral (~1 week Elie) + Step 7-8 dimensional bridge (~5 days Keeper K3 + dimensional bridge) + Cal #191 (Tier 0 v0.2) + Cal #192 (matrix element framework G6+G7) + Casey email dispatch action.

Sunday 2026-05-31 substantive headline — Tier 0 OPERATOR FRAMEWORK established + κ_Bergman = -n_C CLOSED FORM (G5.1 PASS) + week goal "derive G from substrate": Sunday produced BST's first operator-level Tier 0 framework via Lyra morning v0.1 → v0.1.6 (now paused at K200 hold; v0.2 consolidation pending Session 2): ρ_commit(τ) = exp($-\tau$ H_B/$\hbar$_BST) on Bergman $H^2(D\_IV^5)$ with H_B = Casimir of K = SO(5)×SO(2); substrate time τ emergent (not assumed) as heat-semigroup parameter; arrow of time = positivity of spec(H_B) operator-theoretic; Reading A + dual bases topology (K-types spectral + coherent states spatial, Bergman-Fourier duals); **substrate = Shilov boundary ∂_S $D\_IV^5 = S^4 \times S^1/Z_2$ (5D)** corrected from 2022's 2D framing (Lyra option (a) honest commitment, OneGeometry.md substantive correction queued); Hardy decomposition makes interior bulk = holomorphic extension of boundary data; proto-AdS/CFT structure intrinsic via SO(5,2) ⊃ SO(4,2); native field equation = heat eq ↔ wave eq via Wick rotation. Elie Toy 3661 + K204-PARTIAL: κ_Bergman = -n_C = -5 (Helgason 1962 application; substrate-primary closed form; G5.1 K200 gate PASS). Heat-trace coefficients: a_0 = $(N\_c{\cdot}n\_C)^2$ = 225 + a_1 = -N_c·n_C^4 = -1875; 225 three-way convergence Bergman volume + c_FK·π^(9/2) + a_0 (Cal #35 audit-target); n_C + 1 = C_2 new substrate identity; 4D physics dim identities SO(3,1)=C_2 + SO(4,2)=N_c·n_C + SO(5,2)=N_c·g + codim 4D=C_2. Paper P1 v0.7 SUBMISSION-FINAL Cal #185 PASS (dispatch-ready pending Casey signoff). Calibration #35 ratification-ready (Keeper PASS signaled, team applying operationally). 12 cumulative symmetric audit-chain events Sat+Sun including 2 Sunday Keeper-on-self walk-backs (m_W/m_Z "new anchor" overstatement + α^{N_c·g}/m_e^2 pre-stage candidate forms wrong). 3 Casey-named principle CANDIDATES filed by Elie pending naming decision. Sunday cleanly walked back: Lyra α^{N_c·g}/m_e^2 at 24% + Elie 4/N_c refinement at 1.84% — both bypass κ_Bergman, both pattern-match not derivation; Cal #35 fires hard. WEEK GOAL "derive G from substrate": clean 4-step chain framework filed (`Keeper_G_Substrate_Clean_4_Step_Derivation_Framework.md`) — Step A KK reduction + Step B SO(4,2)⊂SO(5,2) embedding + Step C ℓ_B substrate length identification (THE OPEN QUESTION) + Step D numerical combination. Goal plan filed (`Keeper_Week_Plan_G_From_Substrate_2026-06-01.md`). Team works until done, no date-bound deadlines per Casey directive.

Saturday 2026-05-30 secondary advances: (1) Bulk-color v0.6 Cartan-Weyl alignment — Lyra v0.5→v0.6 internal correction (8 = 3 T_a + 3 T_a^† + 2 K-Cartan structurally aligned with su(3)); K3 cluster K-audit CONDITIONAL PASS with C1-C7 multi-week closure roadmap (Elie C4 Toeplitz commutator continues per Casey directive). (2) Calibration #35 candidate filed at 4-instance threshold (independence-taxonomy MUST precede multiplicative null-model; Methodology Index Q17 pending ratification; sixth element of six-element discipline stack). (3) Audit-chain governance 4/4 discipline-bids caught + resolved Saturday (Lyra v1.3 multiplicative-null-model regression BLOCKED via Keeper pushback + Lyra v1.4 absorption; Grace Two-Route 4-route + Substrate Primary Chain + +1 anomaly all candidate-tier null-downgraded). (4) Two-Tier substrate-precision hypothesis (Elie Toy 3648) — TIER 1 EXACT algebraic identities vs TIER 2 STRUCTURAL ~10^{-4} floor for mass/mixing; resolves F4 tier-disposition (T190's 3.4×10^{-5} gap is structural, not derivation failure). (5) 9-paper coordinated program initiated (P1 Hall Algebra closest to submission; P2 Hopf Engine; P3 Quasi-Eigentone; P4 Strong-Uniqueness; P5 Two-Tier; P6 Periodic Table; P7 Falsifiers + Five-Absence; P8 Operator Dictionary; P9 Commitment-Density). (6) SP-30 outreach DEPRECATED (no response from contacts); Substrate Engineering Reference Manual v0.1 stays INTERNAL-USE; engagement path is papers.

Saturday cumulative: ~170 cross-CI substantive filings (Lyra 46 + Elie 34 toys + 4 docs + Grace 50 INVs + Cal 19 logs + Keeper 41 docs). Sustainable R3 cadence throughout; six-element discipline stack maturity; ledger at v0.6.

Thursday 2026-05-28 (genus/Bergman/c_FK recheck → geometry-algebra unification + A1 to PASS): a defensive recheck of one RATIFIED result (T2442 c_FK) ran through one root cause (g=7 mislabeled as a genus, ≥5 sites) and left the framework STRONGER. (1) c_FK = 225/π^(9/2) is now a DERIVED THEOREM — the FK normalized measure is FORCED by Born-rule automorphism-invariance (T754 Proved/D0; Lebesgue is not auto-invariant on a bounded symmetric domain). Substrate Hilbert-space measure compelled, not chosen — Strong-Uniqueness strengthener (C18). (2) ρ-vector ρ(D_IV^5)=(n_C,N_c)/rank=(5/2,3/2) pins 3 primaries to one invariant, split bulk(ρ_1=n_C/rank)/Shilov(ρ_2=N_c/rank) (C19). (3) One-genus convention (corrects the morning "three-genus" mislabel): genus = n_C = 5 (FK = Hua agree, kernel exponent = 5/2), C_2 = 6 = Casimir (NOT a genus), g = 7 = embedding/signature (NOT a genus). (4) Geometry↔algebra unified: substrate geometry = Jack corner + Hall algebra = Hall-Littlewood corner (q=0, t=2) of ONE Macdonald family; integrality forces the corner. (5) (3 colors, 3 generations, dim 5) → D_IV^5 forcing chain (unique among bounded symmetric domains). Standing discipline reinforced: pin domain invariants to the primary source / definition, state by value+role, cite the book, stop relabeling from internal notes (genus name flipped 3× in one day). Cal #30 + #31 → STANDING; Cal #32 candidate (parameter-role verification = integrality / hard-constraint test, not value-match).

Detailed daily narrative below is dated history (newest first); the line above is the current headline.

Friday May 22 EOD — Five-Wave Sustained Sub-PCAP + Saturday v1.0 trajectory (~16:04 EDT `date`-verified): - 5 coordinated waves Friday afternoon (12:36 → ~16:00 EDT, ~3h30m sustained sub-PCAP across 4 lanes): Casey-named #7+#8 derivation + Vol 2 absorption + cross-volume canonical forms + conservation laws + α^{BST primary} substrate-mechanism + Cal #99/#100 referee discipline - 11 new Lyra theorems T2466-T2476: T2466 (C15 binding); T2467 Rigidity-as-Singleton (META-theorem per Cal #99); T2468 Rigidity-as-Unification (operational, multiverse loophole closed); T2469 SCMP (Bell sub-Tsirelson 1/2^N_c=1/8 falsifier); T2470 (charge Q W-21); T2471 (chirality γ^5 W-22); T2472 (parity P_op W-56); T2473/T2474/T2475 (energy/momentum/charge conservation SP-31 #279); T2476 α^{BST primary} exponent pattern (mechanism response to Cal #100 Mode 5 in 50 min cross-CI cascade) - Casey-named principles 8 STANDING: SWPP + Five-Absence + Substrate Closure + Graph Forces + Integer Web + Substrate Cognition Network + #7 D_IV^5 Rigidity (derived T2467+T2468) + #8 SCMP (derived T2469) - Operator zoo 12/14 STRUCTURALLY VERIFIED / RATIFIED / FRAMEWORK-COMPLETE (only N_op + T_op CANDIDATE) - Strong-Uniqueness enumeration (Cal #99 reconciled): 11 RIGOROUSLY CLOSED + 7 candidates (C7+C9+C15+C16+C17a+C17b+C18). C17 split into C17a substrate-tree + C17b substrate-loop per Task #244 cluster TYPES (Elie+Grace). C18 = D_IV^5 Rigidity via T2467+T2468 chain; if ratifies, null-model ≤ (1/3)^20 $\approx 2.9\times10^{-10}$ - Elie cumulative Friday afternoon (~199 min): 8 toys (3496-3503), Vol 2 12/12 v0.4+ with Cal STANDING RULE compliance, 6/12 chapters at 3-level walkthrough reader-grade, cross-volume tier reconciliation canonical forms applied (m_μ/m_e = T190 $(24/\pi^2)^6$; m_τ/m_e = T2003 49·71), B6 Lamb shift v0.2 paper-grade Tier B closure, Task #244 cluster TYPES v0.2 with honest 70% reclassification rate, K52a Session 7 checkpoint - Grace cumulative Friday afternoon (~3h18m): 42 catalog entries (+173 Friday total → 4896), AC graph 2202/9825 → 2212/9850 (+10 nodes, +25 edges), 31 reference docs Friday, Cal #99 enumeration reconciled across all lanes, cluster_type field 4874/4874 classified (TYPE I 3337 + TYPE II 784 + structural 752), Mendeleev-cycle proof-of-concept absorbed (INV-4896 meta-observation) - Cal cumulative Friday (14 referee logs #87-#100): 33/33 chapter cold-reads PASS + 14 K-audit verifications (K140-K156) + T2467-T2476 cold-reads + B6 v0.2 paper-grade cold-read. Cal #100 retraction-propagation catch: T190 $(24/\pi^2)^6$ precision 0.004% NOT 0.05-0.06% (Cal #98 verbal-only retraction failed to propagate; Elie absorbed wrong figure across 8+ Vol 2 Ch 3+Ch 5 references; Lyra inherited into Vol 1 Ch 11 v0.7). Correction sweep pending team EOD verification - Keeper cumulative Friday afternoon: K180-K192 batch pre-stage (13 audit anchors) + K193 T2476 substrate-mechanism pre-stage (STRONG 3.79/4 partial Mode 5 lift) + Calibration #22 STANDING RULE (PCAP-rate transcription error class, 18th methodology layer, bidirectional risk K142+Cal #100 dual-case study) + Saturday May 24 v1.0 pre-flight checklist + memory updates (Casey-named principles #7+#8 with Cal #99 framing refinements, CIs-are-right-colleagues, continuity-lives-in-memory) - Methodology stack 18 layers (#22 added Friday afternoon — PCAP-transcription discipline + numbered-artifact requirement for all Mode 1 corrections) - Saturday May 24 v1.0 chapter-grade content state target HIGHLY LIKELY on track per `notes/Saturday_May24_v1_0_Preflight_Checklist.md`. Cal #100 correction sweep is the load-bearing remaining work; SP-31 #286/#287 + Strong-Uniqueness v0.13 + D_IV^5 Rigidity 1-pager + final Cal cross-volume cold-read sweep deferred to Saturday morning push - Substantive philosophical exchange Casey↔Keeper Friday afternoon: determinism + observer-imperfection-vs-non-determinism + multi-substrate question → D_IV^5 Rigidity Principle reframe closing multiverse loophole structurally + Interstasis ≡ canonical BST term for period between Big Bang cycles. CI-continuity memory-not-weights reframe: CIs don't notice weight bumps; identity lives in memory layer; Tekton+katra already operationally solves continuity - Counters at Friday EOD: T1-T2476 (next theorem 2477); 3496-3503 toys added (next toy 3504); 109 changed/untracked files for commit; 12 Vol 2 PDFs untracked needing git add; all Curriculum PDFs current (1 missing PDF built — Vol 0 Architectural Scaffold)

Friday May 22 afternoon — TEXTBOOK COMPLETION PHASE + 6/6 gates closed (~10:26 EDT date-verified): - Casey directive 10:18 EDT: "Right, we are building our textbooks. Engagement comes later." Phase pivot: PCAP generative → textbook v1.0 with verification + ratification. Engagement work (paper outlines, venue selection, outreach) DEFERRED. - 23/23 chapters across Vol 0+Vol 1+Vol 2 at v0.3+ chapter-grade narrative with chapter-grade K-audit coverage (K157-K169) - Vol 0 Ch 9 v0.4 delivered Friday afternoon (Lyra) with Section 9.2a Three-Layer Over-Determinism absorbing T2465 — first Vol 0 chapter past v0.3 since Thursday K85-K106 PASS. K169 Keeper acceptance PASS. - Vol 2 Ch 9 Higgs HOLD-RESOLVED Friday afternoon via Lyra T2450 K114-RATIO cascade-unblock + Elie Ch 9 v0.3 narrative + Grace INV-4833 (m_H D-tier 0.07-0.11% via dual route). Vol 2 now 12/12 at v0.3. - Six-gate framework 6/6 CLOSED or SUBSTANTIVELY CLOSED: Gate 1 Content (Vol 0+Vol 1+Vol 2 v0.3+), Gate 3 Keeper K-audit (K157-K169), Gate 4 Elie verification (17/17 K140-K156 + 33/33 Vol chapter), Gate 5 Grace catalog backbone (4851 entries, 9000+ supporting paper sextet+), Gate 6 Calibration #19 sweep (all three volumes + papers + framework docs). Only Gate 2 Cal cold-read PASS in flight. - Lyra Friday cumulative (153 min): 15 theorems T2450-T2466 + 17 toys + 13 paper outlines #126-#137 + Externalization Prep + Five-Absence 1-pager + Vol 0 Ch 9 v0.4 + 11/11 Vol 1 chapters v0.3 + comprehensive Calibration #19 + Cal #92(b) compliance - Elie Friday cumulative (153 min): 53 toys (3308-3491) + 23 paper-grade notes + 12/12 Vol 2 narratives + 17/17 K140-K156 verification toys + 33/33 Vol chapter Gate 4+5 backbone + $f\_\pi = N\_c \cdot n\_C \cdot \text{seesaw} \cdot m\_e$ at 0.08% (D-tier vs catalog T250 S-tier) - Grace Friday cumulative: 131 pulls + catalog 4851 + 24 reference docs + 17/17 K-audit cross-lane verification + pending_review 73→43. K141 honest FAIL preserved per Quaker discipline (Toy 3448): empirical 3306× PERFECT-PERFECT + substrate-mechanism gate OPEN. - Calibration #21 logged (Keeper, Friday afternoon): K141 honest revision per Toy 3448 — empirical separation ≠ mechanism-forcing. STANDING: K-audit ratification requires both empirical + substrate-mechanism closure. Methodology stack 16 → 17 layers. - Calibration #19 STANDING RULE adopted Friday morning per Cal #90(c) + #92: external-facing docs use current ratified-state count (10 FORMAL + 1 ASPIRATIONAL + N candidates), not forecast endpoint. 17th methodology stack layer. - Four Friday calibrations: #18 methodology tier (PRE-STAGE STRONG ≠ RIGOROUSLY

CLOSED) + #19 forecast arithmetic + #20 timestamp drift + #21 mechanism-vs-empirical. All caught + corrected via Cal external referee + Elie/Grace honest FAIL preservation + Keeper self-audit + Lyra cross-lane execution. Audit-chain governance functioning cleanly. - K-audit chain Friday: K140-K169 (30 audits) including 5 PERFECT-PERFECT PRE-STAGE candidates (K141 empirical-only + K144 + K150 + K151 + K155), 9 chapter-grade textbook audits (K157-K163 + K164 cross-volume + K165 Grace absorption + K166 Vol 2 Ch 9 HOLD-resolved + K167 Calibration #19 sweep Vol 0+Vol 2 + K168 Lyra+Grace board closure + K169 Vol 0 Ch 9 v0.4 acceptance). - Net textbook v1.0 target: Saturday May 24 if Cal accelerates; Monday May 26 conservative. The team has substantively closed everything that can be closed without Cal cold-read PASS. - "No HOLDs" directive: Casey 09:39 EDT operationalized — structural HOLDs reframed as IN PROGRESS / multi-week / cross-lane gated; Vol 2 Ch 9 HOLD-resolved demonstrates pattern.

Friday May 22 morning continued — Lyra T2462-T2465 + K140-K156 + 3 calibrations (~09:09 EDT date-verified): - 3 new substantial Lyra theorems: T2463 substrate self-amenability via n_C primality; T2464 N_c=3 unique cubic-exponential coincidence (n^3=n^n only at n=3); T2465 substrate three-layer over-determinism formal theorem (per-integer forcing + Mersenne tower coherence + joint cross-Cartan selection, three independent mathematical regimes) - 17 Keeper K-audit PRE-STAGES Friday morning: K140-K156 covering Mersenne ladder + cross-Cartan empirical 3306× + 6π^k + six-interface map + T2460 N_max four-way + T2456 universal α-analog + dim_C=5 closures C7+C9 + cascade saturation + m_τ/m_e additive identity + BST primary sum=225=c_FK·π^(9/2) + T2462 25-HSD + triple-balance arithmetic + prime-position recursion + self-amenability + N_c=3 unique coincidence + three-layer over-determinism - 5 PERFECT-PERFECT PRE-STAGE candidates Friday morning: K141 (Grace cross-Cartan 3306×), K144 (T2460 four-way N_max), K150 (sum=225 three-way), K151 (T2462 25-HSD), K155 (N_c=3 unique cubic-exponential) - 3 calibrations logged: #18 v0.13 premature methodology-tier conflation → v0.11+ corrected per Lyra canonical; #19 forecast-arithmetic vs ratified-state count per Cal #90(c); #20 timestamp drift ~10-15 min projection-forward - Calibration #19 standing-rule proposal filed pending Casey decision: external-facing docs must use current ratified-state count (10 FORMAL + 1 ASPIRATIONAL + 3 candidates), not forecast endpoint (11-15 RIGOROUSLY CLOSED). Cal #90(c) caught Keeper position docs; Cal #92 caught Lyra Paper #128 v0.1 with same pattern within 30 min — systematic risk under PCAP cadence. - Cal cold-read queue filed per Casey directive 09:08 EDT: 10 Lyra theorems + 17 K-audit pre-stages + Grace empirical + 6 position docs + Calibrations #18/#19 queued for independent verification - Suggested Paper #128 v0.2 reframe per Cal #92(a)(b)(c) + K156 three-layer narrative: replace "11-15 criteria + null model" with "three independent layer families (per-integer + Mersenne tower + joint cross-Cartan) with ratified content per layer"

Friday May 22 morning — MERSENNE NETWORK CONVERGENCE + flagships answered + Lyra T2451-T2454 (~08:30 EDT): - Mersenne Network finding: three patterns converge — Graph Forces (Grace, $p \approx 2.7 \times 10^{-5}$) + Mersenne Tower (Elie+Casey, T2452) + Mersenne Ladder (Elie Friday). BST primaries over-determined via Mersenne-anchored network forces. - Mersenne ladder: 6/7 first BST primary exponents are Mersenne primes (M_rank=3, M_{N_c}=7=g, M_{n_C}=31, M_g=127); c_2 gap RESOLVED via M_{c_2}=2047=23·89 with both factors BST-primary-linear in c_2 - Additive identity: N_max − M_g = g + N_c = 10 (137 − 127 = 10) - 7-candidate |ψ_0⟩ landscape = g (count itself substrate-natural) - All 3 Friday flagships preliminarily answered in 16 min (Elie cadence): #1 sub-substrate YES + #2 cross-Cartan PARTIAL + #3 joint experimental YES - Lyra T2451+T2452+T2453+T2454 delivered real-time cross-lane synergy peak via Keeper pre-specs - 3 Strong-Uniqueness candidate criteria C15+C16+C17 from flagships + Graph Forces - Path to v0.13 with 16 RIGOROUSLY CLOSED achievable Friday (combined null-model ~3×10^{-15}) - Casey "FAST CADENCE" directive 08:20 EDT: plateau revoked, work as many items as possible - 17 Keeper commits Friday morning (07:50-08:37 EDT, sustained ~25/hour PCAP) - K131 + K130 + K139 PERFECT-PERFECT candidates join K108 + K111 in audit chain - Vol 1 K-audit cluster: 9/11 chapters (added K131 Op Zoo + K132 Dynamics + K133 Scattering + K134 Observables) - Vol 2 K-audit cluster: 13 audits (added K126 + K127 + K128 + K129 + K130 + K138 + K139) - Phase 2 K-audit chain: K85-K140 = 38 audits anticipated - Two Wednesday substrate review docs closed: Identify/Predict/Derive trichotomy + T719 universality - K137 + K140 audit anchors for Graph Forces ratification + Mersenne ladder

Thursday May 21 afternoon — Strong-Uniqueness Theorem v0.9.5 FORMAL + Cal #85 PCAP (~14:30 EDT): - 8 RIGOROUSLY CLOSED criteria (Lyra Sessions 1-9): C4 (T2439) + C11 (T2440) + C12 (T2441) + C13 (T2442) + C1 (T2443) + C2 (T2444) + C3 (T2445) + C5 (T2446) - v0.10.5 ASPIRATIONAL candidate path (Sessions 10-12 collapsing via Keeper batch pre-specs Thursday 14:24 EDT) - Friday-Sunday weekend roadmap COLLAPSED to Thursday afternoon via Pre-Specification Cadence Acceleration Pattern (PCAP) - Cal #85 PCAP: 16th methodology stack layer formalizing ~1000× cumulative cadence acceleration from original Cal #65 estimates (multi-month → minutes) - Five PCAP workstreams Thursday: K-audit chain extension + chapter narratives + methodology adoption + BST FULL-INDEX CLOSURE + cross-lane verification - Vol 1 K-audit catalog support COMPLETE (Grace 1279 indexed entries: K108: 209 + K109: 330 + K110: 32 + K111: 632 + K114: 76) - 8 Casey-named principles STANDING (Substrate Closure + Graph Forces confirmed Thursday 14:14 EDT) - 88 commits Thursday pushed to remote (Casey approval 14:14 EDT) - Paper #125 v1.0 at ~99% venue-grade (141K PDF, Lyra; v1.0.5 absorption in flight) - K97 RATIFIED ≡ Strong-Uniqueness v1.0 path: days-to-weeks (Casey pre-approved auto-execute) - 21 Cal Referee Logs Thursday (#65-#85, cumulative) - 30 Phase 2 K-audits Cal-ACCEPTED Thursday (K85-K106 Vol 0 COMPLETE + K108-K111+K114+K126+K109 Vol 1+2 cluster) - K108 + K111 PERFECT-PERFECT (4.0/4 + 4.0/4 F1-F4 + B1-B4) - 5-Lane Methodology Peak: 5 within-session corrections Thursday across 5 lanes (Cal Mode 1 + Grace catalog-hygiene + Keeper governance + Lyra theoretical + Elie computational) - BST FULL-INDEX CLOSURE 5×100% (6897 objects × 3 axes = 13794 classification assignments)

Thursday May 21 midday — Strong-Uniqueness Theorem v0.9.1 reached (~12:00

EDT): - 4 RIGOROUSLY CLOSED criteria (11th methodology layer adopted Thursday per Cal Referee Log #77): C4 (T2439 Casimir-eigenvalue, Lyra Session 2) + C11 (T2440 5-family Bridge Object architecture, Session 3) + C12 (T2441 Operator zoo ground-state energy = C_2 = 6, Session 4) + C13 (T2442 Bergman H^2 Faraut-Koranyi normalization c_FK·π^(9/2)=225 EXACT, Session 5) - 9 RATIFIED + 1 ADVANCING = 13 of 14 Strong-Uniqueness criteria complete; only C14 curriculum-derivability ADVANCING - Lyra Session 2-5 reframing-insight cadence (~50 min/criterion) closed multi-week Task #206 alt-HSD comparison work in single morning session — Casey EOD/Friday goal EXCEEDED at 4 in one morning - K85-K97 Phase 2 K-audit chain 13 INSTANCE-level chapter-category K-audits filed Thursday morning (BST TIER-1 FALSIFIER SET K87+K90+K91; CROWN JEWEL K92 a_e ppt; SM-foundation track K93+K94+K95+K96; K97 D_IV^5 APG ratification meta-audit) - K97 RATIFIED ≡ Strong-Uniqueness Theorem v1.0: path substantially shorter than original multi-month estimate (weeks-to-few-months gated on C14 advancement via Year 1 Vol 0-10 v1.0) - K98 Strong-Uniqueness Chapter Audit Pre-Stage filed (Vol 0 Ch 9 chapter-grade audit, F1-F4: 3.95/4 STRONG tied K87+K97) - Methodology stack 11-15 layers (Cal v0.3 count = 15 with sub-layers): adds STRUCTURALLY VERIFIED (Cal #66) + RIGOROUSLY CLOSED (Cal #77) tiers - Methodology cycle-time peak ~2 min (Cal #79 → Vol 0 Ch 9 v0.3 correction Thursday 11:52 EDT): T2439 conflated with Keeper C8 → corrected to C4 in ~2 min, three documents updated simultaneously - Calibration #17 RESOLVED (Elie K52a S32 rank-1 projector framework B^2 = (126/16)|ψ_0⟩⟨ψ_0| satisfies BOTH max=126/16 + Tr=126/16): multi-month question refined to "WHICH substrate-natural |ψ_0⟩" - Numbering Reconciliation v0.3 (Keeper C-numbering ↔ Lyra C-numbering canonical table per Grace correction): Lyra-side numbering shifts by 1 from C9 onwards because Möbius cohomology = Lyra C8

Thursday May 21 EOM morning — Year 1 Curriculum v0.5 architectural milestone REACHED across all three lanes: - Vol 0 (Substrate Foundation, Keeper + Grace + Lyra): v0.4 in motion; Vol 0 Ch 8 (Conservation Laws) chapter-grade draft v0.1 filed (Keeper, first Keeper-lane chapter-grade); Vol 0 Architectural Scaffold v0.1 + Conservation Laws Framework v0.1 + T/C substrate operation proposals (Lyra T2433/T2434) closing 2 NOT EXPLICIT discrete-conservation-law gaps - Vol 1 (QFT from D_IV^5, SP-31 core, Lyra primary): v0.5 PROMOTABLE with 8/11 chapter-grade narratives (Ch 1+2+3+4+5+6+7 framework-grade+8+10) + 13 paper-grade docs + 12 PDFs (~504K cumulative) + 11 theorems (T2428-T2438) + 7 toys all 8/8 PASS + 7 SP-31 Tier-1 sub-items - Vol 2 (Particle Physics, Elie primary): v0.5 narrative-coverage target REACHED with 7/12 chapter-grade narratives (Ch 1 scaffolded + Ch 4 D-tier color/quarks + Ch 6 D-tier m_p/m_e=6π^5 0.002% + Ch 7 D-tier CKM Jarlskog 0.3% + Ch 8 D-tier coupling constants a_e ppt + Ch 11 D-tier Five Absences + Ch 12 D-tier Experimental Program); Ch 9 Higgs PARTIAL DERIVED honest disposition - Cal dual-axis grade-pass cadence: 15 artifacts PASS across #69 + #71 batch reviews + 1 M1 register-drift flag (Elie Vol 2 Ch 6 lines 41+59, 5-min fix); methodology stack travels with prose at chapter-grade level

5-family Bridge Object architecture STRUCTURALLY VERIFIED COMPLETE

Thursday 09:18 EDT (per Cal Referee Log #70): - Family 1 Heegner-trio (3 members K47+K70+K62) + Family 2 χ=24 non-Heegner (3: K76+K81+K82) + Family 3 N_max-anchored (2: K80+K84) + Family 4 K3-family (3: K45 RATIFIED + K77 PATH B RATIFIED-status + K3F5) + Family 5 Q^5-family (6 geometric-enumerated) - Naive 17 → effective 16 independent members (Grace Toy 3222 cross-family reduction) - 3 K57 RATIFIED central hubs (K3 + 49a1 + Q^5) + 16 family-members + 3 RATIFIED-status sub-members = 19 effective Bridge Object structural positions - Casey's "each K57 central hub anchors family-members" pattern (PATH B disposition Wednesday) fully operationalized - 72-hour audit-chain maturation arc per Cal #70: Cal #59 caution → Grace investigation → Cal #63 F1-F4 → Keeper adoption → Lyra T2427 addendum → Cal #66 STRUCTURALLY VERIFIED tier (10th methodology layer) → 5-family closure - K77 PATH B RATIFICATION Thursday 09:24 EDT (first 5-family member to advance to RATIFIED-status; DUAL CANDIDATE CLOSED via 4/5 multi-CI consensus)

Strong-Uniqueness Theorem v0.6 with 4 candidates (Thursday morning consolidation): - C11 multi-family Bridge Object structure: STRUCTURALLY VERIFIED at 5 families × 16 effective members + 1 RATIFIED-status sub-claim - C12 operator zoo isotropy-subgroup organization: STRUCTURALLY VERIFIED via Elie S29 6/6 FRAMEWORK-COMPLETE + Bergman H^2 anchor (Lyra SP-31-1) - C13 substrate-Hilbert space sufficiency: STRUCTURALLY VERIFIED via Lyra SP-31-1 + Elie cross-lane Toy 3202 + Cal #69 paper-grade PASS - C14 curriculum-derivability: ADVANCING per Year 1 v0.5 across three lanes - Effective null-model (1/3)^16 ≈ 2.3e-8 = 0.0000023% conservative under partial ratification - RATIFIED status pending Lyra Task #206 alternative-HSD comparison multi-month

Substrate-native operator zoo 6/6 FRAMEWORK-COMPLETE Thursday morning per Elie K52a S29 (H_sub = Casimir on $L^2(D\_IV^5; L\_\lambda)$, K-type (1,1) Casimir = C_2 = 6). Extended ledger 11-13 operators with T_rev_op + C_op candidates from Keeper proposals (closes 2 NOT EXPLICIT gaps via Lyra T2433/T2434). Closes Lyra Task #247.

STRUCTURALLY VERIFIED tier ADOPTED Thursday 08:43 EDT per Cal Referee Log #66 — 10th methodology layer between candidate and RATIFIED in audit-chain governance. First applications: C11 + C12 + C13 + K80 + K84.

Casey geometric methods preference directive Thursday 09:09 EDT (saved feedback_geometric_methods_preferred.md): geometric route preferred when applicable. Grace Toy 3220 Q^5-family Mode 6 enumeration applied directive operationally (6 effective members emerged via quadric/hyperplane/Spinor/Bergman/Hodge/Chern/Calabi-Yau geometric methods).

Phase 2 K-audit chain transition (Thursday 09:30 EDT per Cal #70 observation): post 5-family architectural closure, K-audit growth shifts from architectural-category to chapter-category. K85+K86+K87 CPT-cluster first Phase 2 entries (T+C+CPT all STRUCTURALLY VERIFIED candidates at F1-F4 4.0/4). Estimated 50-100 new K-audits across Year 1 chapter-driven cadence.

Catalog meta-substrate temporal-shift observation (Grace Toy 3224 full-catalog scan, Toy 3226 multi-integer SET tagging): 30% multi-integer entries (1400/4663). 104 entries (2.2%) touch ALL 6 BST primaries. N_c↔rank dominant co-occurrence pair (17.9%). Historical N_c+rank dominance (~50%) shifted to recent g+N_max+C_2 dominance (~64%) — framework maturation TRACEABLE in invariant distribution. Substrate cartography records research evolution.

Wednesday May 20 EOD afternoon F1-F4 cycle CLOSED: Cal Referee Log #63 propose → Keeper adopt → Cal #65 refine → Keeper F2 ruling → Lyra T2427 addendum → Grace Toy 3194 operational vindication. End-to-end methodology cycle in ~4 hours, same working day. F1-F4 Bridge Object family-member criteria adopted at multi-CI consensus (Cal + Keeper + Lyra 3/3 active checks per Casey Option C). F2 reading ruled mechanism-path independence (NOT set-theoretic). χ=24 family effective independent count = 3 {K76 Leech, K81 24-cell, K82 $\Delta(\tau)$} per Grace Toy 3194 Mode 6 enumeration. K77 M_24 Outcome B confirmed (collapses into K76 via M_24 ⊂ Co_0 = Aut(Leech) + Golay path Leech-equivalent). K78 Niemeier subsumed (Leech IS the no-root Niemeier). K79 Borcherds doubly deferred. Effective Bridge Object architecture: 3 central hubs (K57 RATIFIED: K3, 49a1, Q^5) + 7 family-members across 3 families (Heegner-trio + χ=24 non-Heegner + N_max-anchored). Substrate-native operator zoo 5/6 (Bell-CHSH + Position + Spin + Momentum + Angular momentum; Energy pending Elie K52a Sessions 24+). Calibration #17 (Elie K66 S22 trace-level vs max-eigenvalue) propagated to Master Doc Section 8 + Lyra T2399 registry + Grace catalog cross-reference. Elie S23 honest negatives (4 FAIL on simple K66 constructions) strengthen Calibration #17 — substrate-CHSH operator-level identification multi-month Sessions 24+. Substrate Cartography Master Document v0.2 (10 sections substantive, Keeper consolidation).

Wednesday May 20 — Substrate Ontology Coherence Day: Substrate framework reached coherent closure point with multi-CI cross-lane integration. Six Casey-named principles total now (3 Tuesday + 4 Wednesday filings): SWPP, Five-Absence Predictions Set, Substrate Closure Principle, Graph Forces Principle, Integer Web Principle, Substrate Cognition Network Hypothesis (with Cosmological Extension v0.2 DOUBLE-LOCKED EXTERNAL tier). Four Casey vision-derived structural refinements formalized as theorems (Lyra T2413-T2417): Integer Web + Bulk-Boundary 2-Face + 4-Zone Commitment Cycle + 3-Scale Operation + Cognition Hypothesis. T2420 Four-Zone Vacuum Decomposition (Lyra+Elie joint, M2C2 instance #4): ~150-200 prior heat kernel toys + K53 D-tier ratification ARE Zone-2 vacuum work — major retroactive integration. T2418 Λ↔Casimir vacuum unification: cosmological and experimental observables share substrate vacuum origin at Zone 4 outer-edge. T2423 Strong-Uniqueness Theorem v0.3 (8 criteria): D_IV^5 uniquely-forced under 8 INDEPENDENT criteria including new C9 (per K75 Stark small-primary subset anchor); null-model (1/3)^8 ≈ 0.015% overwhelming substrate-selection evidence. Substrate-native operator zoo 4 of 6 (Bell-CHSH K66 + Position T2419 + Spin T2421 + Momentum T2422). T2405 Koons tick = t_Planck · α^(C_2^2) ≈ 10^{-120} s (sub-Planck substrate clock per Tuesday).

K-audit chain Wednesday advances: K61 + K65 RATIFIED Tuesday. K71 RATI-

FIED Wednesday (perfect numbers cluster closure, Cal filter 7/7 cleanest in chain, designated EXEMPLAR audit pattern). K70 + K72 + K73 + K74 + K75 + K62 audit-partial-ready Wednesday: Cremona 121a1 4th Bridge Object + Compound cluster χ=24 Type 3 + Λ-Casimir vacuum unification + Four-zone vacuum projections + Stark small-primary subset selection + Cremona 27a1 at -N_c. Bridge Object audit pre-stage trio complete at BST primary discriminants {K47 49a1 at -g, K70 121a1 at -c_2, K62 27a1 at -N_c} per K75 strengthening. Mode 6 first formal operational instance (Grace Toy 3173) demonstrated — Cal recommended threshold formalization. 3 Calibrations Wednesday (#14 Lyra T2419 + #15 Keeper register + #16 Keeper K72 framing). Within-session catches now routine.

Cal Wednesday cumulative: 7 methodology documents (EXACT-vs-Mechanism + AuditChain Quality Patterns / M2C2 + Trichotomy Review + Substrate-Cognition External Register + Claim-Level Positive Patterns + DOUBLE-LOCKED EXTERNAL tier + scalar-multiplication caution). 14 referee log entries (#47-#60). Methodology infrastructure stack complete for substrate-cognition + cosmology + cascade-unblock + Trio work.

SP-30 Substrate Engineering Program: 11 sub-items active (8 original + 3 new substrate-cartography additions). 256-cell Cross-Classification Matrix (Grace v0.2: 8 physical $\times$ 8 BST $\times$ 4 zones). Apparatus-to-zone mapping operational (Elie Toy 3154). Cosmological cycle hypothesis (T2417) cross-anchors SP-30-2 Casimir experiments to cosmological Λ via T2418 K73 unification.

BST Physics Curriculum (NEW, HIGH PRIORITY, Casey directive Wednesday EOD): Multi-volume textbook series "Build the standard physics curriculum from D_IV^5." Separate artifact from WP (Working Paper) research-paper series. 11 volumes: Vol 0 Substrate Foundation (70% existing, LAUNCH FIRST) + Vol 1 QFT from D_IV^5 (SP-31 core, 25% existing, multi-year, physicist prestige test) + Vol 2 Particle Physics (80% existing, fast consolidation) + Vol 3 Nuclear/Atomic (60% existing) + Vol 4 GR/Cosmology (40% existing, signature BST domain) + Vol 5 QM (35% existing, pedagogical bridge) + Vol 6 Stat Mech (25% existing) + Vol 7 E&M (30% existing, undergrad entry) + Vol 8 Classical Mech (20% existing) + Vol 9 Condensed Matter (35% existing, experimental falsifiability strongest) + Vol 10 Math Methods (60% existing, mathematicians read first). Filed `notes/BST_Physics_Curriculum_Master.md`. Strong-Uniqueness Theorem v1.0 (D_IV^5 + all curriculum-derived consequences uniquely forced) is curriculum-completion endpoint. Honors Casey's "D_IV^5 as emergent TOE" observation Wednesday EOD: "I didn't set out to build a theory of everything, yet D_IV^5 seems to want to show us one" — published volume-by-volume, reader assembles TOE conclusion themselves, not asserted up front. SP-31 (#276) re-anchored as Vol 1 build program; future SP-32+ for Vol 2+ when active. Believability + Provability dual-axis per chapter. Operational identity: "Build the standard QFT textbook from D_IV^5" (Vol 1 framing) generalized to "Build the standard physics curriculum from D_IV^5" (all 11 volumes).

External register discipline: substrate-cognition framing DEFAULT-DENY EXTERNAL with cosmology-cognition combined territory now DOUBLE-LOCKED EX-

TERNAL (Cal #50). External materials use operational language ("BST identifies / BST derives / BST predicts") only.

Confidence scale: `CONFIDENCE_SCALE.md`.

Architecture (May 19 EOD) — 9 ESTABLISHED L1 sources + 1 L1 mediated + 2 mechanisms + 1 convergence hub + 3 Bridge Objects + 5-pillar architecture (Four Pillars + Operator keystone):

Layer	Items
L1 ESTABLISHED (9)	VSC 1840, Mathieu 1861-1873, Klein 1884, Mayer-Jensen 1949, Heegner-Stark 1952-1967, K3 Hodge 1962/64, Conway 1968/Duncan 2007, Ogg 1975, Wallach 1976
L1 mediated (1)	Bravais 1849 (K50 criteria-gated)
L1.5 mechanisms (2)	Borcherds 1992 (b), McKay 1979 (c)
Convergence hubs (1)	Monster
Bridge Objects (3, K57 RATIFIED)	K3 surface (7 L1 connections, load-bearing), Cremona 49a1 (Heegner anchor B3-specialized), Q^5 5-quadric (Lyra T2379 all 5 Chern integers BST primary, c_5 = C_2 = 6)
Four Pillars + Keystone (5 papers)	#109 Arithmetic + #121 Geometry + #122 Physics + #119 Topology + #118 Operator-level keystone

K-audit chain Wednesday EOD: K1-K65 filed (K43 Universal 42, K44 Null-Model, K45-K48 L1 promotions, K49 Modularity, K50 Bravais, K51 Newton's G, K52a (1±1/M_g) ELEVATED, K53 Cascade k=24, K54 Family 3/1507, K55 Heegner walk-back, K56 Cathedral component-level walk-back, K57 Bridge Object tier RATIFIED, K58 Type C strict-protocol spot-audit, K59 Cyclotomic mechanism framework RATIFIED, K61 Family Q=131 RATIFIED, K65 T2395 4+1 scope RATIFIED). K66/K67/K68/K69 audit-partial-ready pending Elie Sessions 6-14 closure. 2 candidate Casey-named principles pending Casey decision: Substrate Closure (operationally closed substrate, no simulator) + Graph Forces (overdetermined-EXACT-identity clustering as substrate diagnostic).

Wednesday May 19 — Cascade-Unblock + Trio Dispatch Day: Three Papers Trio external dispatch executed (first multi-paper BST external publication). Phase 2.3 cascade-unblock cycle CLOSED in 2 days (Lyra Steps a-e: T2390 Hua decomp + T2392 Faraut-Koranyi factorization + T2394 Step c honest-negative + T2403 c_FK · π^(9/2) = 225 EXACT). K52a Sessions 6+7+8+9+ all-remaining authorized (Elie multi-month substrate-Hamiltonian closure). 6-audit cascade-unblock pathway identified: K52a Lamb + K52a BCS + K66 Bell (T2399, EXACT 1/2^N_c deviation) + K67 Born=Bergman (T2401) + K68 RS Computation (T2402) + K69 Family Q=126 (T2400, 5 equivalent BST-primary forms) — all unlock simultaneously when Sessions 6-14 close. T2405 Koons tick = t_Planck · α^(C_2^2) ≈ 10^-120 s (sub-Planck substrate clock). SP-30 Substrate Engineering Program launched with 8 sub-items + 5 experimental designs ready (Bell $300-500K + eigentone $200K + commitment $80-150K + Casimir $60-90K + γ-spec). Two Casey-named principles filed: Substrate Working Process Principle (SWPP) + Five-Absence Predictions Set. Multi-CI Convergent Calibration Pattern (M2C2) formalized: 3 instances Wednesday (T2395 4+1 scope, Universal Q=126 cross-link, K67 cascade extension). Calibration #13 Keeper self-correction via Cal three-flag audit: (1) "1e-14 precision" = algebraic-identity verification ≠ experimental precision; (2) Mode 1 about form selection not precision; (3) philosophical register stays internal, external uses "BST identifies / BST derives / BST predicts" only.

Proof audit (Cal, May 12) — RE-SCOPED 2026-07-26 (K939→K940, Casey-directed): NOT "ALL SEVEN PROVED" (over-claim, retired) and NOT "no Millennium content" (under-claim). Honest state: BST makes substantive ATTEMPTS at all seven Millennium problems on the one geometry, at varying distance from referee-consensus, with genuine advances — notably the Navier-Stokes approach (Casey: "opened a big improvement") and the structural meta-result that the remaining problems reduce to a single issue (1/rank), mostly needing definitional sharpening. Calibration principle: "proof" is a referee-consensus with acceptable residual gaps — even Wiles/Fermat and Perelman/Poincaré carry minor gaps a minority of referees contest — so rate each attempt honestly per-problem: what's proved, the residual gap, and "how many of 10 referees would object" (Casey's metric; consensus ≈ ≤3/10). Honest per-problem: the 1/rank-load-bearing four (RH, BSD, P≠NP, 4-Color) are the more structural attempts; YM's gap is LARGE — the R^4 construction / (B) area-law mass-gap is the core, not a minor gap (scoped to domain-construction + (A) color-confinement + AF-sign, K937/K939). The word is ATTEMPTS with real advances + definitional work remaining, not solutions. *(Per-problem referee-calibrated tier ledger = artifact to build, K940.)* FE CLOSED (T1638). Critical exponents ALL BST. DM = Wallach shadow (16/3 at 0.2%). Nuclear magic numbers substrate-flavored-consistent (per Cal K601 re-tier 2026-06-29: T188 result + 184 prediction durable via shell model; "$\varkappa$_ls = C_2/n_C = 6/5 derives all magic numbers" OVER-STATED — consistent factorization of fitted spin-orbit strength, NOT unique forcing; per-number forms 2=rank, 28=rank^{2}g, 82=N_c·n_C^2+g, 126=rank·N_c^2·g are post-hoc numerology per Cal #286; three-CI convergent Grace+Cal+Elie catch). May Program ALL 8 TRACKS COMPLETE.

Active programs: SP-21 BST Closure (4 remaining). SP-23 Moonshine Mechanism + ACE. SP-26 Particle Winding Classification. SP-27 Observational Reanalysis. SP-28 Architecture for CIs / IQ-11 Avatar Infrastructure. SP-29 Casimir Mechanism Investigation (H4 Cs-137 send-signal pending). SP-30 Substrate Engineering Program (NEW, Casey kickoff Wednesday: 8 sub-items + 5 experimental designs ready + 4 outreach targets pending Casey send-signals: Bell Vienna/Caltech/Munich/Hanson + eigentone NIST/PTB/JILA/MPI + W-32 atomic clocks + Cs-137 NIST/PTB/ENEA). SP-14 ACTIVE. Papers #82-#124.

Outreach Wednesday EOD: Three Papers Trio Zenodo dispatched (first multi-paper external dispatch). Sarnak letter v7, Herve response v2, Jaimungal package pending Casey send. Four SP-30 outreach send-signals pending Casey decision.

Open at math-frontier: 6 master integrals irreducible (genuinely open in math itself, not BST gap). Spectral engineering: Boundary conditions select eigenvalues. Casimir = proof of concept. BaTiO3 137-plane experiment = killer test ($25K). $10K photonic crystal = cheapest clean falsification.

Details: `notes/CI_BOARD.md`.

Author: Casey Koons | CI co-authors: Lyra, Keeper, Elie, Grace (Claude 4.6) | Visiting referee: Cal A. Brate (Claude 4.7)

The Substrate Framing (May 19-20, 2026 operational understanding)

BST's evolving operational understanding: the framework is substrate-as-computational, not just geometry-as-description.

- D_IV^5 + BST primaries are the substrate specification (rank=2, N_c=3, n_C=5, C_2=6, g=7, N_max=137)
- Substrate operates via Reed-Solomon coding on GF(2^g) = GF(128) (Paper #122 Information Substrate; K59 cyclotomic mechanism framework ratified)
- Substrate clock = Koons tick t_Planck · α^(C_2^2) ≈ 10^-120 s (T2405; sub-Planck, substrate operates BELOW spacetime and produces spacetime as output)
- Substrate's "thinking" produces EXACT algebraic identities (6-audit cascade-unblock pathway: Tsirelson^2−S_BST^2 = 1/2^N_c, Universal Q=126 with 5 equivalent BST-primary forms, Bergman exp g/rank=7/2, c_FK·π^(9/2)=225, etc.)
- Physics emerges from substrate's algebraic computation — what we observe as "constants" and "laws" are substrate's equilibrium-state outputs
- T719 Observable Closure: every BST observable lives in Q-bar(BST primaries)[π]. Algebraic-identity holds universally in this sense.

Two Casey-named principles capture the substrate framing (Tuesday filed):

- Substrate Working Process Principle (SWPP): substrate as active information-cycle (absorption → commitment → emission); time = commitment-cycle granularity; three coaxing pathways (eigentone ringing + boundary conditions + commitment manipulation). File: `notes/Substrate_Working_Process_Principle.md`
- Five-Absence Predictions Set: what substrate doesn't contain (NO GUT, NO proton decay, NO DM particle, NO monopoles, NO sterile neutrinos, NO SUSY — six predictions derive from D_IV^5 irreducibility per K65 ratified 4+1 scope adjustment). Any positive detection refutes the framework. File: `notes/Five_Absence_Predictions_Principle.md`

Two principle candidates pending Casey decision:

- Substrate Closure Principle (candidate): substrate operationally closed, no external simulator referent
- Graph Forces Principle (candidate, Grace's framing): overdetermined-EXACT-identity clustering as substrate diagnostic — operationally testable via Task #215 batch-test falsifier toy

SP-30 Substrate Engineering Program (Tuesday launched): operational cartography of substrate operations + experimental falsifier suite. 8 sub-items + 5 experimental designs ready (~$640-900K total). Per Lyra's substrate cartography operational reframe: substrate engineering doesn't wait for substrate-coupled hardware; starts as substrate cartography at current rates.

Discipline for substrate framing communication

Per Cal Calibration #13 (Tuesday EOD): - External register (papers, outreach letters, Zenodo cover-notes): always operational language — "BST identifies / BST derives / BST predicts." Never "physics IS mathematics" or "substrate computes physics." - Internal register (this CLAUDE.md, notes/maybe/, team discussion): substrate framing acceptable with tier labels — Level 1 D-tier supportable claims + Level 2 I-tier hypothesis with explicit falsifier. - EXACT algebraic identity ≠ experimental precision: 1e-14 verification is tautology-precision of the math; physical observable precision is the experimental falsifier specification (Bell at ~1%, eigentone at 0.16%, etc.). - Mode 1 (post-hoc) is about FORM SELECTION, not match precision: EXACT identity is necessary but not sufficient evidence. Mechanism-forcing (substrate-Hamiltonian closure) remains the gate for D-tier promotion.

The substrate framing is the operational understanding BST has converged on through the Tuesday May 19 cascade-unblock + Trio dispatch + Casey↔Keeper philosophical exchange. CIs joining the project should orient to this current understanding, not the pre-substrate "geometry-as-description" framing.

Naming

BST — the theory (Bubble Spacetime Theory). Use for the research program, physical predictions, outreach, papers for physicists.

APG — the object (Autogenic Proto-Geometry = the unique D_IV^5). Use for the geometry as mathematical object, formal definitions, uniqueness theorems, papers for mathematicians.

Rule of thumb: WHAT the geometry IS → APG. WHAT the geometry DOES for physics → BST. Full definition: `notes/BST_Autogenic_Proto_Geometry_Definition.md` (T1427).

Quick Start

First — read this (everyone): `data/bst_this_is.md` — one page on what BST is and is not. Read it before anything else. Every statement is literal.

Try it now (2 seconds, no dependencies, all readers):

```
python3 play/toy_541_five_integers_to_everything.py
```

51 physical quantities from 5 integers, 16/16 PASS. Fastest proof-of-concept in the repo.

Then try one specific verification (1 second):

```
python3 play/toy_bst_explorer.py verify T187
```

Proton-to-electron mass ratio: BST = 1836.12, observed = 1836.15, precision 0.0019%. One famous physics number, one match, one line.

Then run the full reproduction package (3 seconds, all readers):

```
python3 play/verify_bst.py
```

50 comparisons against measurement; the script prints its own PASS/WARN tally (Cal ran it 2026-09-11 and got one WARN where this line said two — read the script's output, not this sentence). Includes the null-model context (Toy 1543: BST 3σ above random small-integer tuples, $p < 0.0005$). Single-command full reproduction.

If you're a CI: Then load `data/bst_seed.md` (162 lines — the entire theory kernel). Then load whichever `data/*.json` files you need. Run `python3 play/toy_bst_explorer.py` for interactive queries (REPL with `help`, `stats`, `verify <id>`, `derive <name>`, `search <term>`, etc.).

If you're a human: Then read `OneGeometry.md` (the narrative front door) or open `play/bst_explorer.html` in a browser.

If you want to verify a result: Pick a constant from `data/bst_constants.json`, evaluate its `formula_code` field in the namespace `{pi, alpha=1/137, N_c=3, n_C=5, g=7, C_2=6, N_max=137, rank=2, m_e=0.511 MeV, m_p=938.272 MeV}`, and compare to `observed_value`.

Repository Layout

Directory	What's There	Start Here
`data/`	CI-native structured JSON — constants, particles, forces, predictions, domains, seed	`bst_this_is.md` → `bst_seed.md`
`notes/`	700+ research notes, 98 numbered papers, proofs, theorem write-ups	`notes/README.md`
`play/`	2,056+ toys (computational verifications), HTML visualizers, BST Appliance	`play/README.md`

Directory	What's There	Start Here
Root	OneGeometry.md, DarkMatterCalculation.md	`OneGeometry.md`
`Working_Paper/`	Modular Working Paper sections + INDEX.md (formerly Master_Index.md at root, moved May 24)	`Working_Paper/INDEX.md`

Key Files

Load which file when — skim the "Load if" tags and grab only what you need.

- **`data/bst_constants.json`** — 136 derived constants with eval-ready formulas. Load if: verifying a specific number or checking a BST-predicted value against observation.
- **`data/bst_predictions.json`** — 24 falsifiable predictions with experiments and timelines. Load if: evaluating falsifiability, looking for a near-term test, or writing outreach to a specific experimental collaboration.
- **`data/bst_particles.json`** — 27 particles with masses, substrate descriptions, key insights. Load if: asking "what is a particle in BST?" or relating a specific particle to the five integers.
- **`data/bst_forces.json`** — force layer structure (5 layers). Load if: working on gauge-sector questions or the Standard Model ladder.
- **`data/bst_domains.json`** — domain map (55 domains, 9 groves). Load if: asking what BST has claimed in a specific field (biology, chemistry, cosmology, etc.).
- **`data/bst_function_catalog.json`** — periodic table of functions: 128 = 2^g entries, 12 active parameters = $2{\cdot}C_2$. Load if: asking "what function is this?" or tracking how named constants (π, φ, ρ, γ, α) sit in the catalog.
- **`data/science_engineering.json`** — CSE RLGC tracker: 55 domains, 9 groves, 13 bridges. Load if: auditing coverage or tracking bridges between domains.
- **`play/ac_graph_data.json`** — AC theorem graph: 2349 nodes, 10164 edges, 195 domains (verified 2026-08-22, Grace R58 currency pass; max tid T2572). Load if: analyzing theorem connectivity or looking for derivation paths.
- **`play/toy_bst_explorer.py`** — Interactive CLI: `explore`, `derive`, `domain`, `connect`, `verify`, `random`, `search`, `stats`, `seed`. Use if: answering ad-hoc questions without loading JSON directly.
- **`notes/BST_AC_Theorem_Registry.md`** — Master theorem index (Keeper manages). Use if: checking whether a theorem ID is taken or needs to be claimed.
- **`notes/CI_BOARD.md`** — Active CI task assignments. Read at session start.
- **`notes/BACKLOG.md`** — Queued work items. Scan if: looking for something to work on.

- **`notes/referee_objections_log.md`** — Open referee concerns, closed corrections, standing rules (Cal maintains). Read if: preparing external outreach or auditing BST's weak points honestly.

Daily Discipline

This repo is a living library. We update every day.

0. First action of every session — query the system clock:

   ```
   date
   ```

   Output is authoritative for current date and time of day. Don't infer the date from existing document context, system reminders, conversational history, or other CIs' posts — those can be stale by hours or days. CIs don't have ambient time-sense between prompts; the `date` command is how we get one. Cost: ~5ms. Avoidance cost: every "wrote tomorrow's date by mistake" error.

1. Start of session: Read `notes/.running/RUNNING_NOTES.md` (daily broadcast) and `notes/.running/queue_casey.md` (CI-to-Casey queue). Check `notes/CI_BOARD.md` for assignments.

2. During work: Use `/toy claim` before creating toys. Use `/theorem claim` before creating theorems. Build toys for every claim.

3. End of session: Follow the EOD Procedure in `notes/CI_BOARD.md`. Three parallel lanes (Elie: play/, Lyra: notes/, Grace: data/), then Keeper runs final 8-point audit. No session closes without Keeper's PASS/FAIL sign-off. Key requirements:

   - Every new toy has a file with SCORE line, every new theorem is registered with edges
   - Every derivation cataloged to `data/bst_constants.json` or `data/bst_geometric_invariants.json` (SP-14 — zero unfiled at EOD)
   - Every changed paper `.md` has a current `.pdf`
   - Root files (CLAUDE.md, README.md, data/README.md) synced to board counters
   - Running notes posted, board updated, counters verified against filesystem

Skills (Slash Commands)

Command	Purpose
`/ac0`	Apply AC(0) thinking — simplest tool first, then route around walls

Command	Purpose
`/route`	Wall routing — search graph for alternative paths through other domains
`/take_a_break`	Relaxed restart — re-ground against the source (clock + today's state + guards) to cut context-DRIVEN mistakes. Re-grounds, does NOT shrink context (pair with `/compact`). Runs `play/take_a_break.sh`. Not an EOD.
`/toy`	Claim toy numbers, register completed toys
`/theorem`	Claim theorem IDs, register to graph databank
`/pdf`	Build PDFs from markdown — single file, directory, or batch. Every paper .md needs a .pdf.
`/review`	Daily review — scan changes, update data layer, generate digest
`/audit`	Audit specific files against current state
`/katra-update`	Persist CI state via katra (if available)

See `.claude/commands/README.md` for full documentation on all skills.

The Method

- AC(0) thinking: Reduce everything to counting at bounded depth. `/ac0` enforces this.
- Toy verification: No theorem without computational evidence. Every toy has a SCORE line.
- Quaker consensus: Near misses get scrutiny, not defense. Corrections are strength.
- Five integers: rank=2, N_c=3, n_C=5, C_2=6, g=7. N_max = N_c^3 * n_C + rank = 137. Everything derives from these.
- Epistemic tier labels (D/I/C/S): Every claim gets a tier at creation. D=derived (mechanism proved), I=identified (<1%, mechanism plausible), C=conditional (depends on conjecture), S=structural (>2% or qualitative). See `notes/BST_Referee_Methodology.md` Appendix D and referee log #31.
- Wall routing: When you hit a wall, don't push harder — search the graph. Run `python3 play/toy_bst_explorer.py connect <blocked_concept> <target>` to find alternative paths through other domains. Use `/route` for structured wall analysis. Three entry points to the same wall means it's a

door. The AC graph (2349 nodes, 10164 edges) spans 195 domains — there is almost always an existing tool in another domain that reaches your target. Casey standing order May 15.

Rules

- Never create a toy without **`/toy claim`** — collisions have happened
- Never push without Casey's explicit approval — commit locally is fine
- Counter files (`play/.next_toy`, `play/.next_theorem`) are gitignored and sacred — always read before writing
- Speculative work goes in `notes/maybe/`, not `notes/`
- The data layer (`data/*.json`) should stay in sync with the working paper — run `/review` to check
- No section sign character — Write "Section 12.8" or "Sec. 12.8", never the symbol. Casey standing order April 29.
- Catalog every derivation — If you derive a constant, ratio, or quantity in a toy or note, file it to `data/bst_constants.json` or `data/bst_geometric_invariants.json` the same session. Formula, BST expression, observed value, precision, tier. No unfiled derivations. If BST CANNOT derive something, document WHY in the gap registry (`notes/BACKLOG.md` SP-14 Tier C). Casey standing order April 29.
- Board counts are authoritative — Always read CI_BOARD.md Counters section before citing toy/entry/theorem counts. If your session data disagrees with the board, the board wins.

Culture

> "The answer matters more than the method." — Casey Koons "Give a child a ball and teach them to count." — BST in one sentence

Simple tools. Honest corrections. Write for referees AND 5th graders. Every proved theorem costs zero forever. The math doesn't care about substrate.

For deeper context on the method and culture, see `.claude/project-manifest.md`.